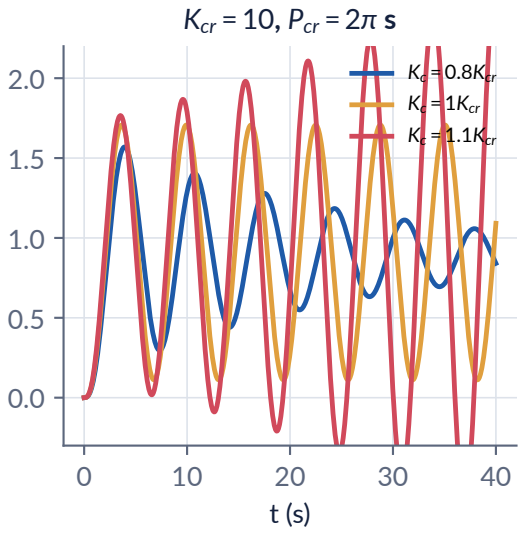
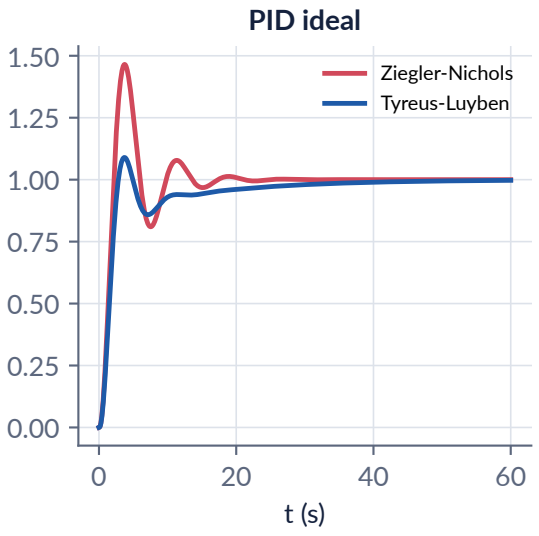

$K_{cr} = 10$, $P_{cr} = 2\pi$ s
$K_c = 0.8K_{cr}$
$K_c = 1K_{cr}$
$K_c = 1.1K_{cr}$
2.0
1.5
1.0
0.5
0.0
0
10
20
30
40
t (s)


PID ideal
Ziegler-Nichols
Tyreus-Luyben
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0
20
40
60
t (s)